

1. In the lecture videos you saw that vectors are linearly dependent if it is possible to write one vector as a linear combination of the others. For example, the vectors $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are linearly dependent if $\mathbf{a} = q_1 \mathbf{b} + q_2 \mathbf{c}$ where q_1 and q_2 are scalars.

1 point

Are the following vectors linearly dependent?

$$\mathbf{a} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}.$$

- ☒ Yes
☐ No

2. We say that two vectors are linearly independent if they are *not* linearly dependent, that is, we cannot write one of the vectors as a linear combination of the others. Be careful not to mix the two definitions up!

1 point

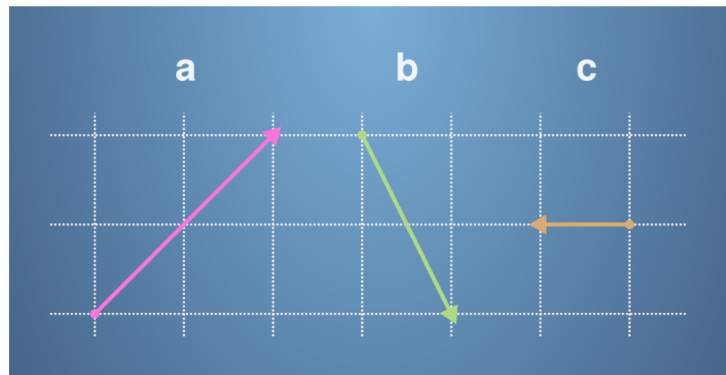
Are the following vectors linearly independent?

$$\mathbf{a} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \text{ and } \mathbf{b} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

- ☒ Yes
☐ No

3. We also saw in the lectures that three vectors that lie in the same two dimensional plane must be linearly dependent. This tells us that $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ are linearly dependent in the following diagram:

1 point



What are the values of q_1 and q_2 that allow us to write $\mathbf{a} = q_1 \mathbf{b} + q_2 \mathbf{c}$? Put your answer in the following codeblock:

```
1 # Assign the correct values for q1 and q2 to write a as a linear combination of b and c
2 q1 = -1
3 q2 = -3
```

4. In fact, an n -dimensional space can have as many as n linearly independent vectors. The following three vectors are three dimensional, which means that we must check if they are linearly dependent or independent.

1 point

Are the following vectors linearly independent?

$$\mathbf{a} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \text{ and } \mathbf{c} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

- ☒ Yes
☐ No

5. Are the following vectors linearly independent?

1 point

$$\mathbf{a} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} \text{ and } \mathbf{c} = \begin{bmatrix} -3 \\ 1 \\ -2 \end{bmatrix}.$$

- ☐ Yes
☒ No

6. The following set of vectors cannot be used as a basis for a three dimensional space. Why?

1 point

$$\mathbf{a} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} -2 \\ 1 \\ 3 \end{bmatrix} \text{ and } \mathbf{c} = \begin{bmatrix} 4 \\ 3 \\ -3 \end{bmatrix}.$$

$$\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \quad \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix} \quad \begin{bmatrix} 3 \\ 6 \\ 9 \end{bmatrix}$$

- ☐ The vectors are linearly independent
- ☒ The vectors are not linearly independent
- ☒ The vectors do not span three dimensional space
- ☐ There are too many vectors for a three dimensional basis